

ASSIGNMENT 5

Vehicle Fuel Efficiency Data Visualization using Python

Aim:

To analyze and visualize vehicle fuel efficiency data using Python libraries such as Matplotlib and Seaborn, enabling better understanding of fuel consumption trends, engine size effects, transmission types, and key factors affecting miles-per-gallon (MPG) performance.

Objectives:

1. To learn data visualization techniques in Python.
2. To represent vehicle fuel efficiency data graphically.
3. To identify patterns and relationships in automotive/fuel data.
4. To create different types of statistical and categorical plots.

Algorithm:

1. Import required libraries such as Pandas, Matplotlib, and Seaborn.
2. Create a dataset containing vehicle fuel efficiency records.
3. Convert the dataset into a Pandas DataFrame.
4. Calculate average MPG and efficiency scores.
5. Generate different types of visualizations: Line Plot, Scatter Plot, Histogram, Density Plot, Bar Chart, Pie Chart, Subplots, Count Plot, Heatmap, Pair Plot, and 3D Plot.
6. Add titles, labels, legends, and grid styles.
7. Display all graphs using `plt.show()`.
8. Analyze the results obtained from the visualizations.

Program:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from mpl_toolkits.mplot3d import Axes3D

# Dataset Matrix setup
data = {
    "Vehicle_ID": ["V001", "V002", "V003", "V004", "V005", "V006", "V007", "V008", "V009", "V010"],
    "Brand": ["Toyota", "Ford", "Honda", "BMW", "Tesla", "Hyundai", "Chevrolet", "Nissan", "Kia", "Mazda"],
```

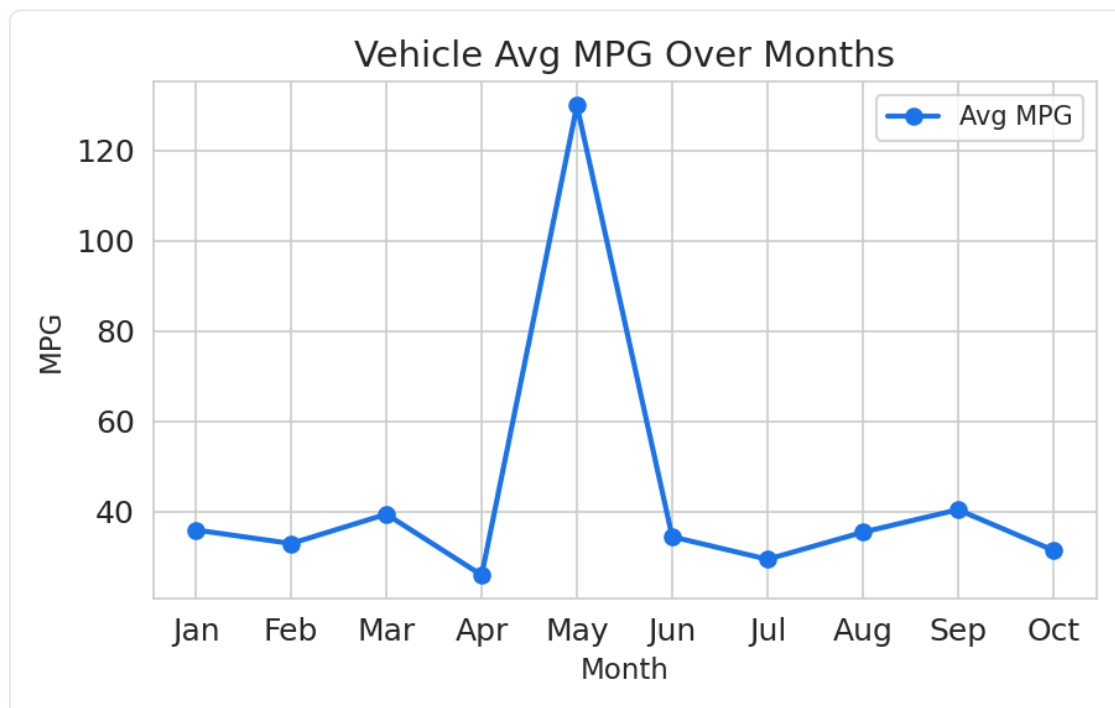
```

    "Fuel_Type": ["Petrol", "Diesel", "Petrol", "Diesel", "Electric", "Petrol", "Diesel", "Petrol", "Petrol", "Diesel"],
    "Transmission": ["Auto", "Manual", "Auto", "Auto", "Auto", "Manual", "Manual", "Auto", "Manual", "Auto"],
    "Month": ["Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug", "Sep", "Oct"],
    "Engine_Size": [1.8, 2.0, 1.5, 3.0, 0.0, 1.6, 2.5, 1.8, 1.4, 2.2],
    "City_MPG": [32, 28, 35, 22, 120, 30, 25, 31, 36, 27],
    "Highway_MPG": [40, 38, 44, 30, 140, 39, 34, 40, 45, 36],
    "CO2_Emission": [120, 140, 110, 180, 0, 125, 155, 118, 108, 145],
    "Mileage_km": [15000, 22000, 18000, 30000, 25000, 12000, 28000, 17000, 9000, 20000],
    "Horsepower": [132, 150, 119, 248, 346, 130, 180, 128, 100, 165],
    "Safety_Rating": [4, 3, 5, 4, 5, 4, 3, 4, 5, 4]
}

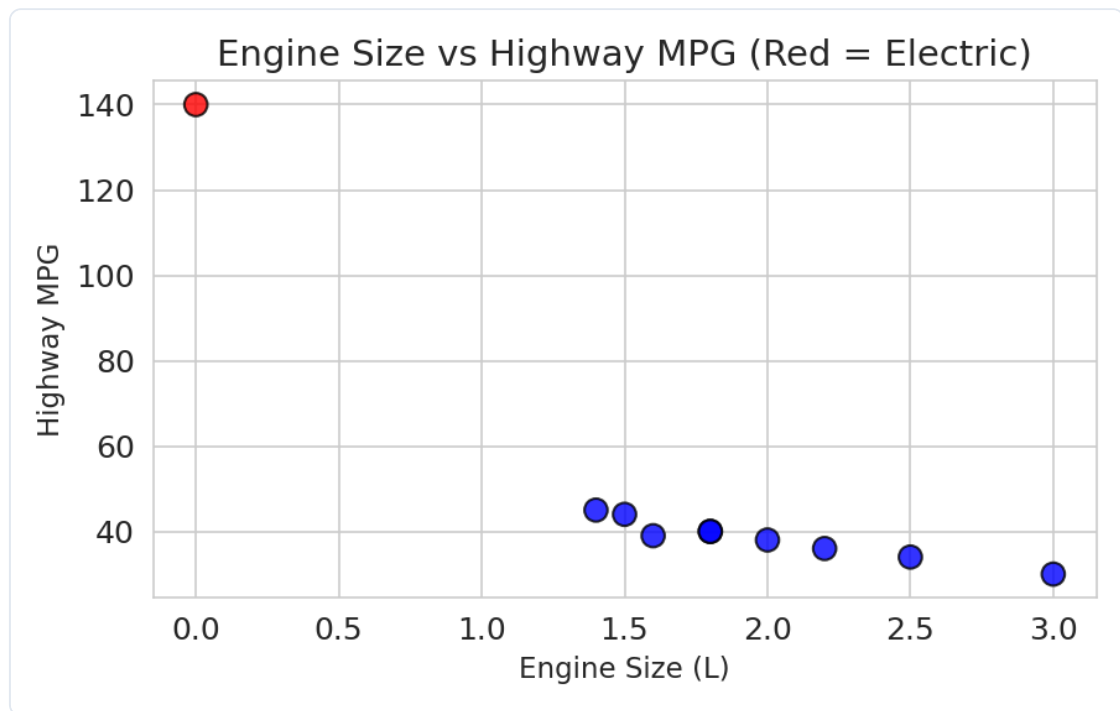
df = pd.DataFrame(data)
df["Avg_MPG"] = (df["City_MPG"] + df["Highway_MPG"]) / 2
sns.set_style("whitegrid")

```

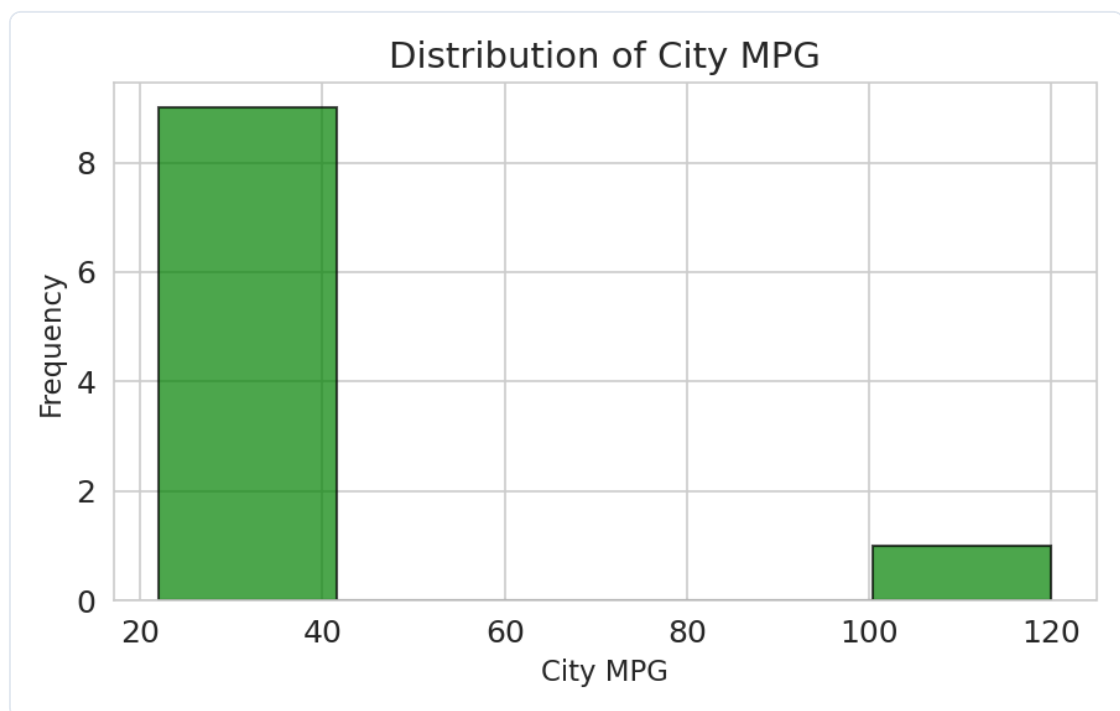
4. Rendered Visualization Plots



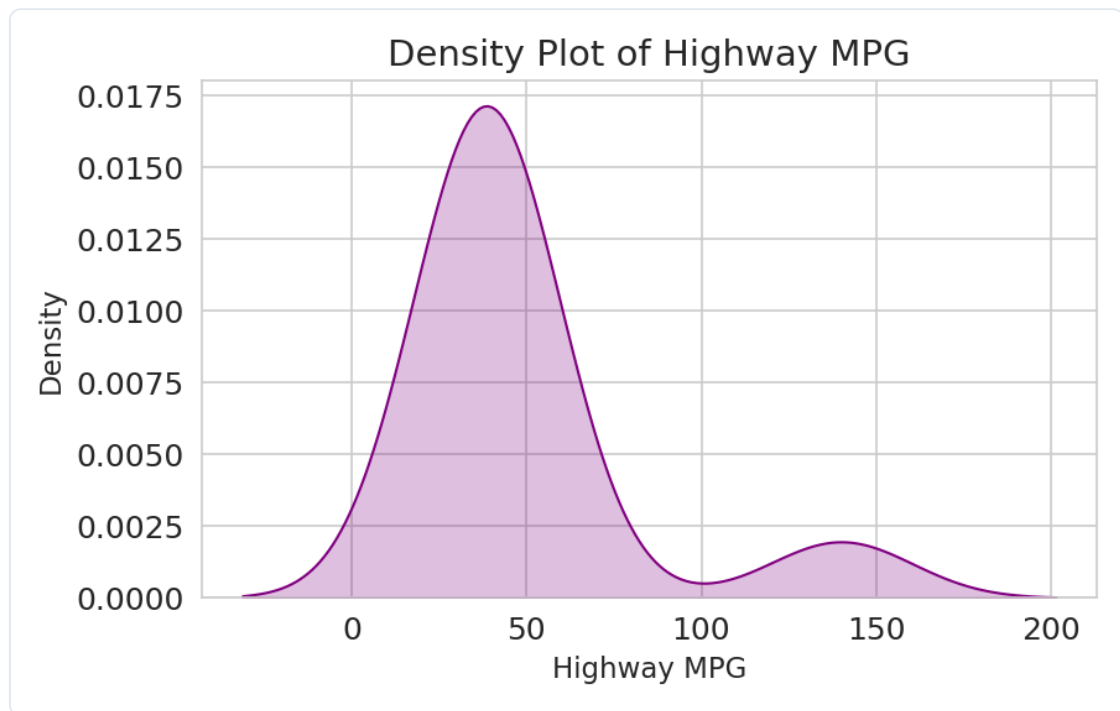
Plot 1: Line Plot - Vehicle Average MPG Over Months



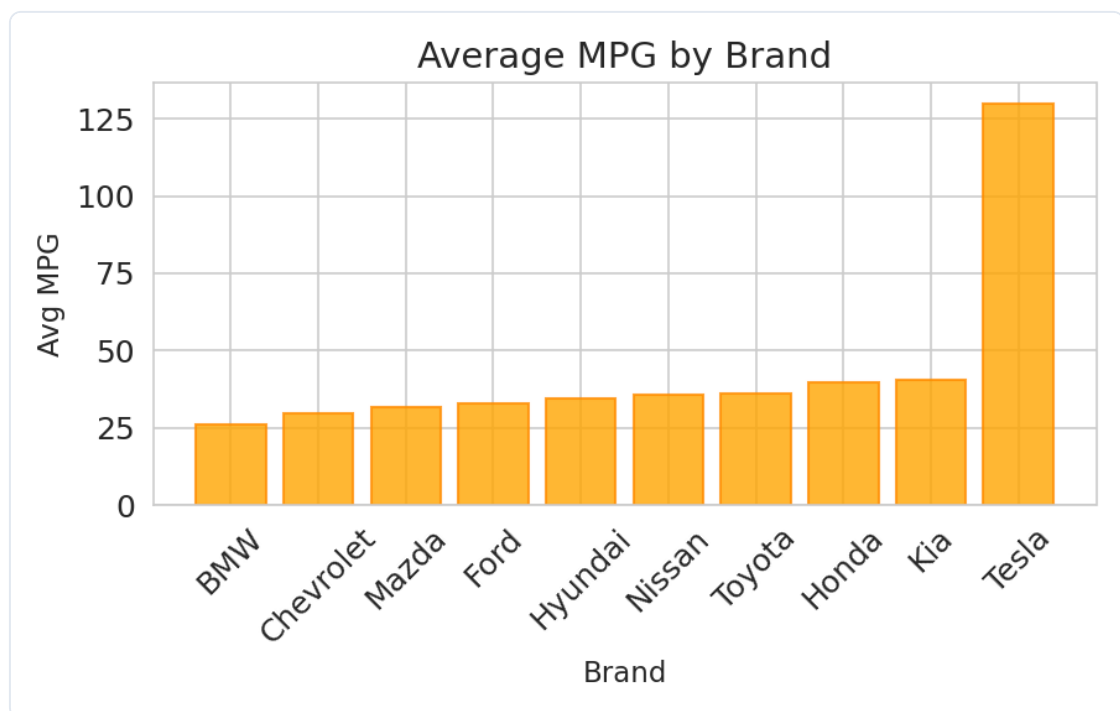
Plot 2: Scatter Plot - Engine Size vs Highway MPG



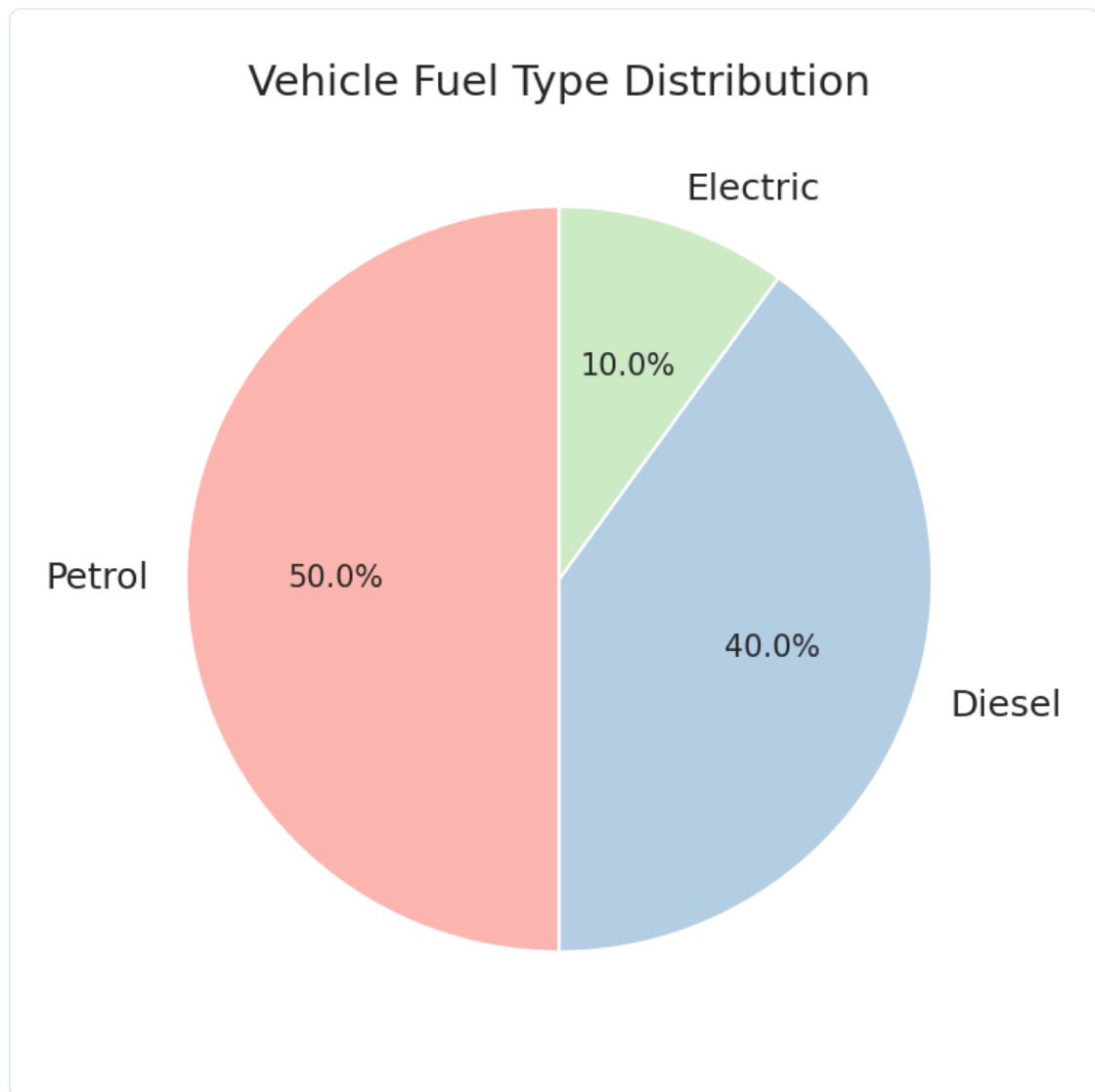
Plot 3: Histogram - Distribution Density of City MPG



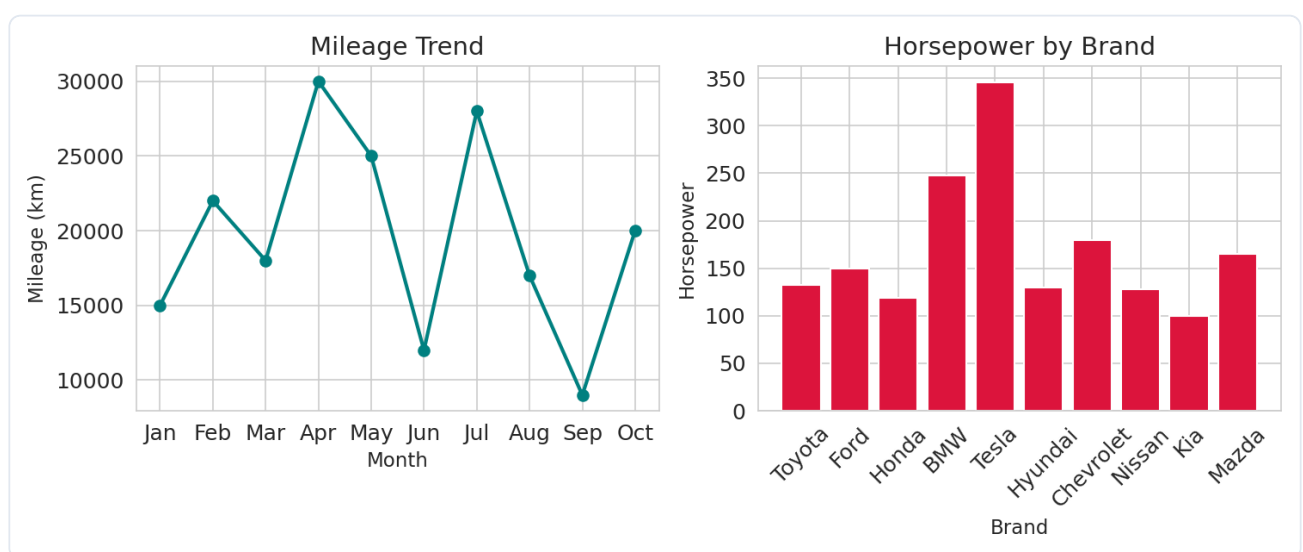
Plot 4: Density Plot - Probability Profile of Highway MPG



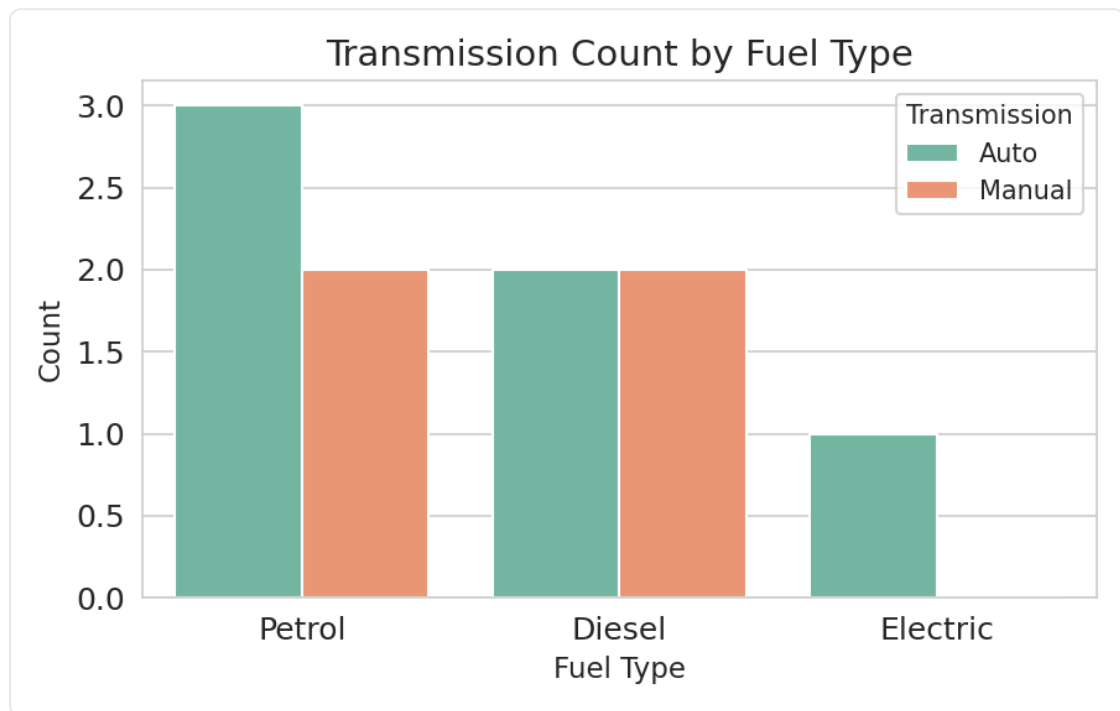
Plot 5: Bar Chart - Average MPG Compared across Automotive Brands



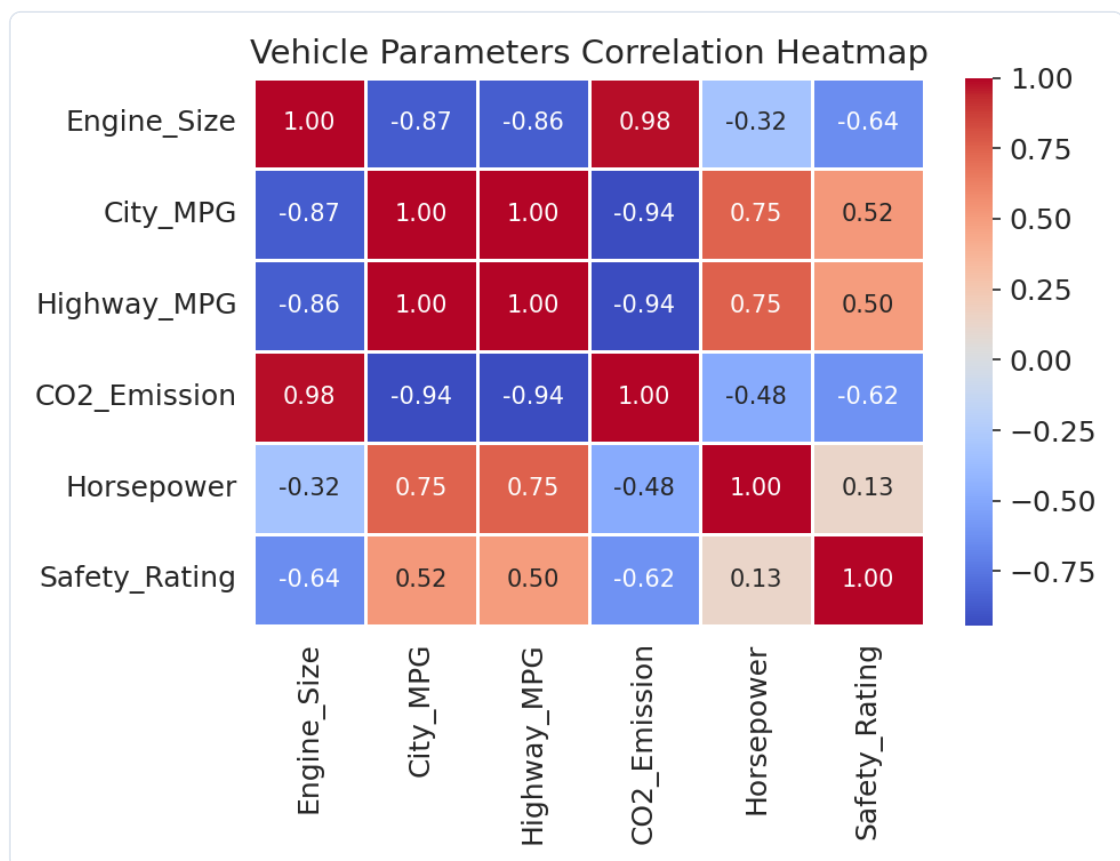
Plot 6: Pie Chart - Structural Vehicle Fuel Type Distribution Share



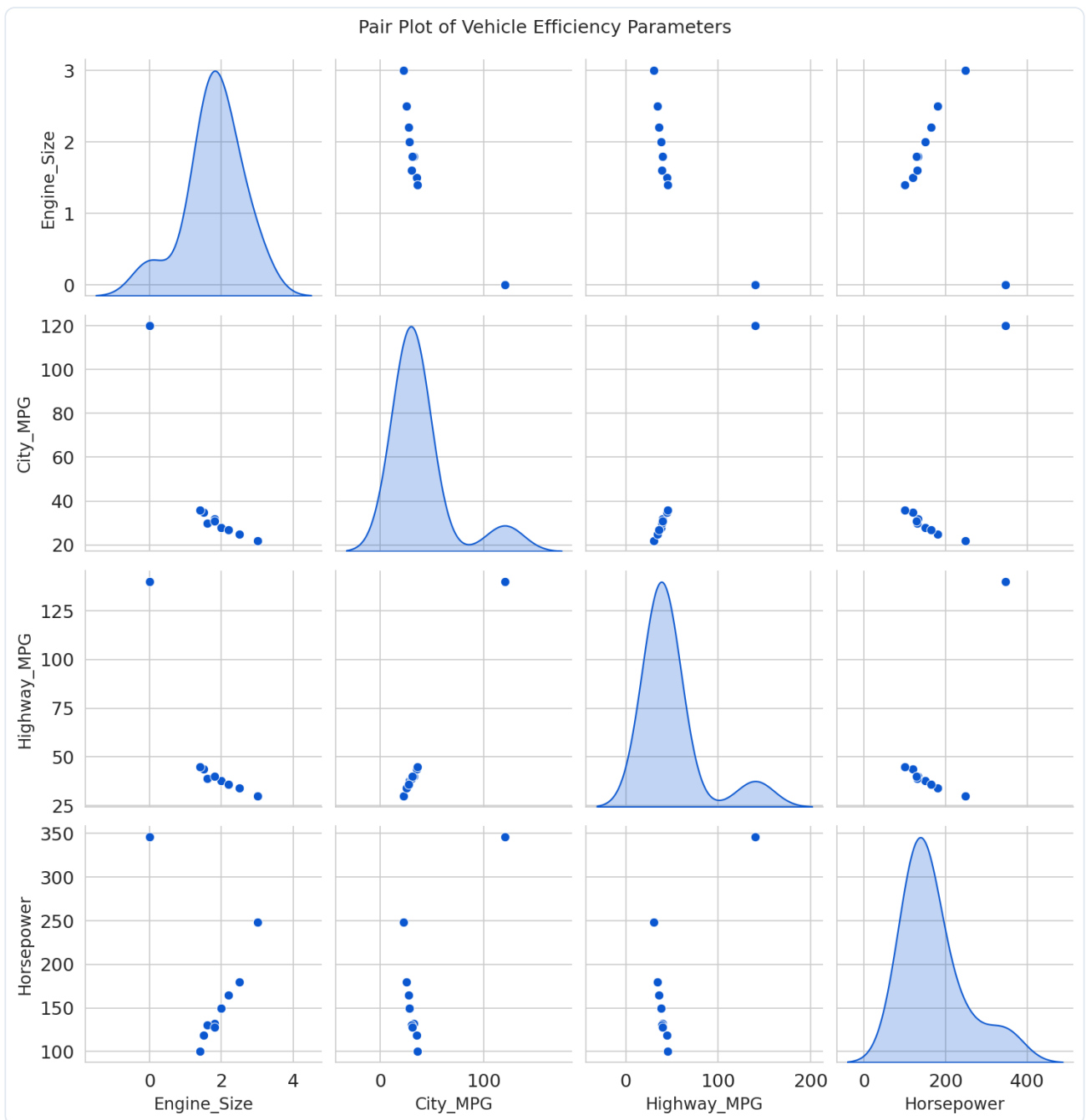
Plot 7: Subplots - Multi-Panel Layout tracking Mileage Trend & Horsepower



Plot 8: Count Plot - Transmission Categorization Grouped by Fuel Type

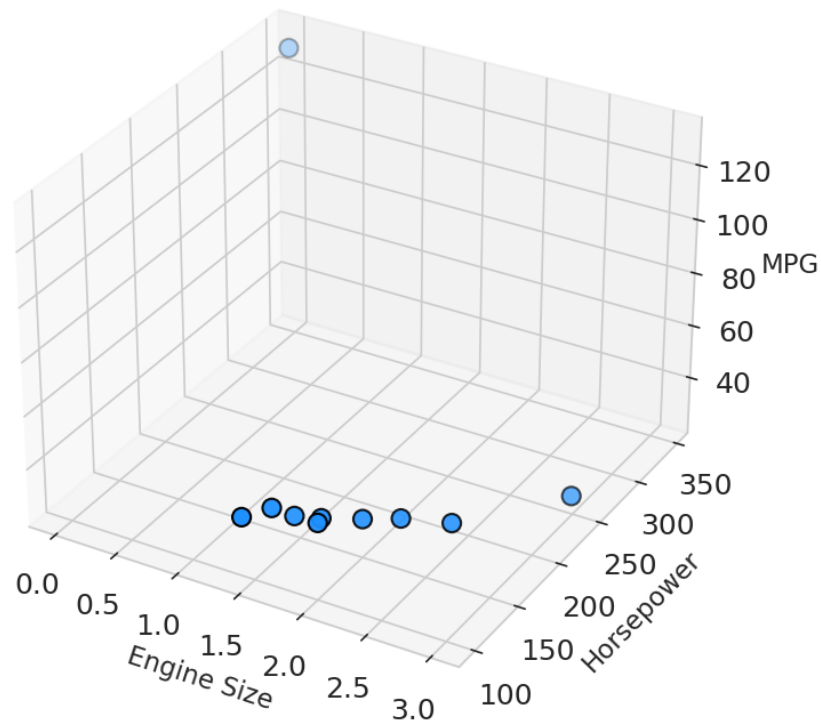


Plot 9: Heatmap - Vehicle Parameters Statistical Linear Correlation Matrix



Plot 10: Pair Plot - High-Dimensional Pairwise Variable Distribution Grid

3D Vehicle Fuel Efficiency Visualization



Plot 11: 3D Coordinate Scatter Plot - Engine Size vs Horsepower vs MPG Analysis

Result:

Thus, the vehicle fuel efficiency data visualization was successfully performed using Matplotlib and Seaborn. Different plots helped in understanding distributions, comparisons, and relationships between vehicle parameters such as Engine Size, City MPG, Highway MPG, Horsepower, CO2 Emissions, and Mileage effectively. The visualizations clearly identified key factors contributing to vehicle fuel efficiency across different fuel types and brands.

Course	B.E (Computer Science and Engineering)
Academic Year	2025-2026 [Even Semester]
Title	Vehicle Fuel Efficiency Analysis
Student Name	Aswanth.P
Register Number	2117250020044
Department	Computer Science Engineering
Institution	Rajalakshmi Institute of Technology
Faculty Name	Dr. H. Anwar Basha
Date	25/05/2026